Arya Hosseini

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SUMMARY

Game Programmer proficient in working with **Unreal Engine**, **Unity** and custom engines. Experienced in developing **gameplay systems**, **tools**, and **engine features** with a strong foundation in game architecture, engine design, and optimization.

SKILLS

Engines & Tools: Unreal Engine, Unity, SFML, ImGui, Git, Tracy

Programming Languages: C++,C# , Lua

PROFESSIONAL EXPERIENCE

Unity Gameplay Programmer Intern

Baobab Games | Montreal, Canada | May 2024 - Aug 2024

- Prototyped and integrated 18 new modular building parts into the **base building system**, each with designer defined constraints and unique gameplay interactions, expanding available building options for players by over 50%.
- Built a world **item storage system**, integrated with the existing UI and player inventory, and optimized with a memory pool and custom physics behaviours, delivering a stable, performant feature that shipped to playtests.
- Collaborated with artists, designers, and QA across sprints in an **Agile** environment to diagnose and resolve gameplay issues.

PERSONAL PROJECT

ECS Game Engine (C++ 20)

View Project

- Built a data oriented 2D game engine from scratch with a custom archetype based **ECS** and CPU friendly memory layout, applying **SIMD** and **cache coherent** data structuring principles, making large scale entity iteration significantly faster.
- Wrote a type erased **memory allocator** for contiguous component storage and recycling, backed by a global component registry for safe cross thread memory operations, eliminating fragmentation and heap overhead in performance critical paths.
- Developed a fully interactive Dear **ImGui editor** and **engine tools** including an entity inspector, rendering viewport and a real time log window with stack trace tracking, enabling scene authoring and debugging without touching source code.
- Created a **JSON prefab system** for persistent entity serialization across scenes and engine lifetime with full editor side mutation support, allowing creating and modifying entities entirely through the GUI.
- Implemented a **rendering system** using SFML's module on OpenGL. Scaled this module by batch rendering of sprite textures via manual vertex arrays, reducing draw calls to the minimum and keeping rendering cost flat as entity count scales.
- Built a **physics pipeline** using a spatial grid to narrow collision candidates efficiently, calculated collision contacts and resolved them with an impulse based approach. Resulting in stable collisions across all geometry configurations.

Action Roguelike (Unity)

View Project

- Designed a **modular player architecture** using the State Pattern with independent handler components for movement, combat, animation, SFX, and VFX, enabling new gameplay states and abilities to be added without modifying existing systems.
- Built a **procedural world generation** system using zone based architecture and binary space partitioning with proximity based culling and coroutine batched generation, ensuring seamless exploration regardless of world size.
- Replaced Unity's built in NavMesh with a custom **Flow Field Pathfinding** system designed for dynamic procedural environments, enabling efficient simultaneous navigation for multiple agents across world layouts that change at runtime.
- Implemented an **Enemy AI** framework with ScriptableObject defined behaviours, producing reusable, scalable AI states that can be configured and extended.

EDUCATION

DEC – Computer Science, Video Game Programming — LaSalle College | Montreal, Canada | 2022 - 2024